

Atrial fibrillation density as a biomarker for ischaemic stroke risk prediction

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Abstract

Background and Aims

Atrial fibrillation (AF) is frequently classified by episode duration and cumulative burden, yet these methods fail to characterize temporal episode distribution (AF density), which may be more clinically relevant and prognostically important. This study evaluated AF density’s association with ischaemic stroke and whether it improves risk stratification compared with AF burden.

Methods

Data from two US cohorts with cardiac implantable electronic devices remotely monitored by the Veterans Health Administration and University of North Carolina (January 2010–May 2025) were analysed. AF burden (percentage of time in AF) and density [range: 0 (dispersed episodes) to 1 (consolidated episodes)] were assessed in 30-day intervals and categorized as low (>0–0.3), medium (>0.3–0.6), medium-high (>0.6–0.9), and high (>0.9–1.0). Patients with permanent AF or no episodes ≥ 6 min were excluded. G-formula modelling estimated 1-year stroke risk ratios (RRs), adjusting for baseline and time-varying covariates. Results were pooled using random-effects meta-analysis.

Results

Of 41 780 patients, 12 868 met inclusion criteria (mean age 72.0 years; median CHA₂DS₂-VASc 4.0); 336 experienced ischaemic stroke over a median 4.0-year follow-up (6.3 per 1000 person-years). Atrial fibrillation density demonstrated a dose–response relationship with 1-year stroke risk (RR 1.75; 95% confidence interval 1.25–2.44) and findings were consistent across device types, comorbidities, age, and anticoagulation status. At each level of AF burden, patients with high density exhibited greater stroke risk.

Conclusions

Atrial fibrillation density demonstrated robust dose–response relationships with ischaemic stroke independent of AF burden and enhanced risk stratification, suggesting that density may enable more precise stroke risk assessment and personalized prevention strategies.

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Structured Graphical Abstract

Key Question

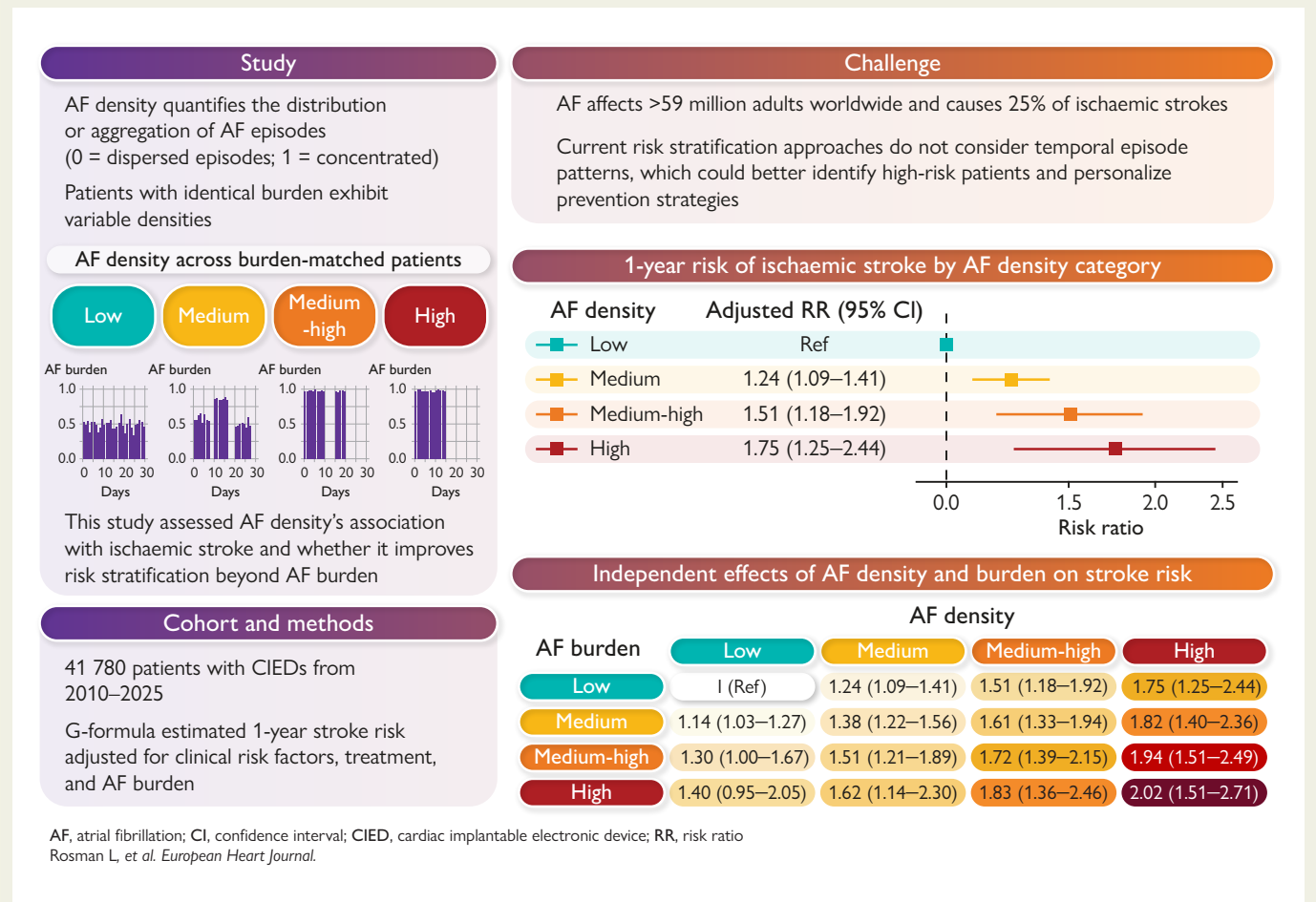
Does temporal clustering of atrial fibrillation (AF) episodes (AF density) improve stroke risk stratification beyond current measures of AF burden?

Key Finding

In this large, multicentre analysis of U.S. patients with implanted cardiac devices, AF density demonstrated a robust dose–response relationship with ischaemic stroke independent of AF burden and established risk factors. Results were consistent across device types, comorbidities, age groups, and anticoagulation status.

Take Home Message

AF density is a novel stroke risk metric demonstrating that temporal clustering of AF episodes improves risk stratification beyond cumulative arrhythmia burden.



variability in AF with corresponding fluctuations in stroke risk.⁵ Studies of AF burden (the proportion of time spent in AF during a monitoring period) generally show a dose–response relationship with stroke, with higher burden associated with increased risk.^{4–8} This relationship is supported by large-scale rhythm control trials, including ATHENA and EAST-AFNET 4, which demonstrated that interventions reducing AF burden lower stroke rates.^{9,10} Nevertheless, important questions remain about the clinical utility of AF burden. The burden–stroke relationship is non-linear, and no consensus exists regarding burden thresholds that should guide clinical management. For example, the ASSERT trial found stroke risk increased with AF episodes exceeding 24 h,¹¹ whereas the SOS-AF project demonstrated significant associations at >5 min.¹² These inconsistent thresholds suggest that quantifying AF as a single time-based percentage may be insufficient to capture the complexity of arrhythmia patterns relevant to stroke risk.

Atrial fibrillation density is another potentially important metric for quantifying the distribution or aggregation of AF episodes during a monitoring period.¹³ Values range from 0 to 1 with scores closer to 0 representing a low density (AF episodes spread evenly throughout monitoring period) and scores closer to 1 reflecting a high density (AF episodes clustered together in blocks or continuous episodes). Prior studies have demonstrated that patients can have equivalent burdens but variable densities.¹³ Whether density is clinically relevant and prognostically important remains unknown. Accordingly, we leveraged continuous remote monitoring data from 41 780 US patients with cardiac implantable electronic devices (CIEDs) from two distinct and diverse health systems to evaluate the association between AF density and risk for ischaemic stroke. We also examined whether AF density improves stroke risk stratification compared to AF burden.

Methods

Study design and data sources

We conducted a multicentre, retrospective observational study using data from two US registries of patients aged ≥18 years with CIEDs [pacemakers, implantable cardiac defibrillators (ICDs), cardiac resynchronization therapy devices, and insertable cardiac monitors (ICMs)] who were remotely monitored by the Veterans Health Administration (VHA) and the University of North Carolina (UNC) Health System from 1 January 2010 to 30 May 2025. The VHA is a large, single-payer, integrated US health system with a centralized remote monitoring programme serving 122 clinics nationwide. Patients are automatically enrolled in the VHA Cardiac Device Registry upon initiating remote monitoring. The second cohort included patients with CIEDs implanted at UNC, an 11-hospital academic health system in central North Carolina, who were automatically enrolled in the UNC Cardiovascular Device Surveillance Registry (UNC CDSR) upon initiating remote monitoring.^{14,15}

Both registries prospectively collect comprehensive CIED data, including arrhythmia episodes and daily physiological data [e.g. physical activity (min/day) and AF burden (h/day)] from remote monitoring transmissions and device interrogations for all CIED implants, upgrades, and replacements at participating centres. Device data are deterministically linked to comprehensive clinical information, including demographics, comorbidities, medications, procedures, and outcomes, which are routinely abstracted from

electronic health records (EHR) using standard procedures and validated International Classification of Diseases, Ninth and Tenth Revision (ICD-9/10) codes and Current Procedural Terminology (CPT) codes.¹⁶ Each registry also captures unique supplemental data: the VHA includes laboratory results, vital signs, and pharmacy dispensing records, while UNC includes patient-reported lifestyle factors (i.e. alcohol consumption). To ensure comprehensive outcome ascertainment in the VHA cohort, data from the Centers for Medicare and Medicaid Services, VA Community Care, and the National Death Index were linked to the VHA registry to capture clinical events occurring outside that health system. Comorbidities and outcomes were identified using validated ICD-9 and ICD-10 diagnostic codes.¹⁶ Cause-specific mortality data were obtained from the VA Mortality Data Repository and National Death Index (VHA cohort) and North Carolina State Death records (UNC cohort). Detailed information regarding data sources and methods for each cohort is provided in the [Supplementary data online, Methods](#) (see [Supplementary material](#)).

Study population

The study population included all eligible patients from the participating health systems meeting inclusion criteria: patients aged 18 years and older who received their first dual-chamber ICD, pacemaker or ICM implanted during the study period and had a daily AF burden ≥6 min post-implantation.¹⁷ Patients with single-chamber ICDs and pacemakers without atrial sensing, >10% missing data during the follow-up period, permanent AF, left ventricular assist devices, or heart transplantation were excluded from the analysis.

Definitions of atrial fibrillation, burden, and density

Atrial fibrillation detection during follow-up was based on CIED manufacturer algorithms that have been extensively validated, demonstrating accuracy of 98%, 100%, and 98.5% for AF detection in dual-chamber pacemakers and ICDs, and ICMs, respectively.^{18–22} Daily AF burden was defined as the total minutes of AF per day that met CIED AF detection thresholds. Only patients with at least 1 day of total AF ≥6 min during follow-up were included, as this threshold was associated with increased stroke risk in the SOS-AF study.¹² Intracardiac electrograms were not available to adjudicate AF episodes. For patients with ≤10% missing remote monitoring data, daily AF burden was imputed using the last value carried forward.

Atrial fibrillation density quantifies the temporal clustering of AF during each 30-day monitoring period and is defined as the ratio of cumulative deviation of actual AF burden from a hypothetical uniform distribution to the maximum possible aggregation ([Supplementary data online, Methods](#)).¹³ Values range from 0 (brief, dispersed, disconnected AF episodes or low density) to 1 (consolidated, uninterrupted AF episodes or high density).

Atrial fibrillation burden was calculated as the proportion of total time in AF during each 30-day monitoring period. Atrial fibrillation density and burden were both transformed as a proportion and analysed on the same 0–1 scale to enable direct comparison.

To assess dose–response relationships and enhance clinical interpretability, AF burden and density were assessed in 30-day intervals and categorized as low (>0–0.3), medium (>0.3–0.6), medium-high (>0.6–0.9), and high (>0.9–1.0). This categorization was chosen to (i) enable direct head-to-head comparison of burden vs density on an identical 0–1 scale, (ii) ensure adequate sample sizes across strata given the distribution of CIED-detected AF in our cohorts, and (iii) align with prior studies²³ that examined associations between AF burden and outcomes.

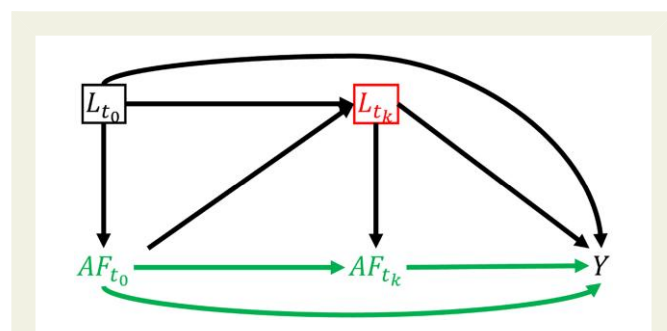


Figure 1 Overview of the g-formula approach for analysing time-varying atrial fibrillation (AF) density and stroke risk. This directed acyclic graph illustrates the temporal relationships between AF density (AF, green text), time-varying confounders (L, boxed variables), and ischaemic stroke outcome (Y), measured at baseline (t_0) and subsequent intervals (t_k). Arrows represent causal pathways. Baseline confounders (L_{t_0} , black box) include patient characteristics such as age, sex, and comorbidities. Time-varying confounders (L_{t_k} , red box) include additional factors such as physical activity, blood pressure, and anticoagulation status

Baseline comorbidities and time-varying covariates

Potential confounders and effect modifiers were selected *a priori* based on clinical knowledge and prior literature and incorporated into a directed acyclic graph (DAG) to guide the modelling strategy (Figure 1; Supplementary data online, Methods). The DAG illustrates the causal structure between time-varying exposures, time-varying covariates, and ischaemic stroke risk.

Baseline characteristics included demographics, comorbidities, procedures, lifestyle factors, and medications, while time-varying covariates (see Supplementary data online, Tables S2–S4) were measured at the start of each person-month to ensure they temporally preceded the exposure and outcome. These included lagged AF burden and density, body mass index (BMI), systolic blood pressure, ejection fraction, CIED-detected daily physical activity, and medications such as anticoagulation therapy [prescription claims and/or dispensed quantities of oral anticoagulation (OAC), including warfarin, apixaban, edoxaban, dabigatran, or rivaroxaban]. The CHA₂DS₂-VASc score was calculated for each patient at device implantation. Missing data were coded as 'Unknown' for categorical variables, or the medians for other continuous variables.

Outcome

The primary outcome of interest was the first occurrence of ischaemic stroke after CIED implant. Ischaemic stroke events were identified using validated ICD-9/10 diagnostic codes (see Supplementary data online, Table S1) during hospitalizations (ICD-9: 433.X1, 434.X1, 436; ICD-10: I63), or ischaemic stroke as the underlying cause of death (ICD-10: I63, I64, I69.3, I69.4). These ICD-9/10 codes have been extensively validated in VHA data, demonstrating a positive predictive value of 90%–100% in prior studies.²⁴

Patients were followed from the date of CIED implantation until the occurrence of an ischaemic stroke event, death, or last trend date from the device, whichever happened first. This study followed the STROBE reporting guidelines.²⁵ Institutional review boards at each site approved the study protocol with waiver of informed consent. Device manufacturers provided no funding for the study. All authors assume responsibility for data integrity and analyses.

Statistical analysis

Baseline characteristics of both cohorts were summarized as frequencies and percentages, means and standard deviations (SDs) for symmetrically distributed variables, or medians and interquartile ranges (IQRs) for skewed variables and stratified by those with and without a clinical diagnosis of AF at the time of implant. Data management and statistical analyses were conducted using SQL Server Management Studio 19.0 and R version 4.3.1.

Since AF density varies over time and that clinical factors influencing stroke risk (e.g. physical activity and blood pressure) may themselves be affected by prior AF episodes, we used the parametric g-formula²⁶ to estimate ischaemic stroke risk. Technical details of this statistical approach are provided in the [Supplementary data online, Methods](#). This method extends traditional approaches by accommodating exposures and confounders that vary over time and are causally interrelated.^{27,28} For example, a patient's physical activity level affects both their AF density and stroke risk, but activity itself often declines after experiencing AF episodes.²⁹ Conventional Cox models cannot properly account for such bidirectional relationships and produce biased estimates. The g-formula addresses this by appropriately modelling how AF density and risk factors change together over time to estimate the true effect of AF density on stroke risk.

Briefly, for each person-month from baseline through follow-up, we fit regression models for all time-varying covariates (e.g. AF burden, AF density, CIED-measured physical activity), for the risk of non-stroke death (as a competing event), and for the risk of ischaemic stroke. Both AF burden and density were modelled as continuous variables. We then simulated a cohort in which AF density was fixed to the specified categories, while baseline covariates were set to their observed values. Competing risks were accounted for via a sub-distribution hazard approach to directly estimate the cumulative incidence function. Using the fitted models, we sequentially predicted the values of time-varying covariates and the probabilities of ischaemic stroke and non-stroke death at each subsequent time point under these hypothetical AF density assignments. This process was repeated recursively across all time points, and the final 12-month ischaemic stroke risk was obtained by averaging the predicted risks across all individuals. The 95% confidence intervals (CIs) (2.5th and 97.5th percentiles of the bootstrap distribution) for cumulative incidences, risk differences, and risk ratios (RRs) were estimated using 500 bootstrap samples.

All models adjusted for baseline confounders including device type, implantation year, age, sex, race, tobacco use, BMI, clinical diagnosis of AF or flutter, CHA₂DS₂-VASc score, alcohol or substance use disorders, chronic kidney disease, cardiomyopathy, diabetes, depression, dyslipidaemia, heart failure, hypertension, obstructive sleep apnoea, valvular heart disease, history of ischaemic stroke or transient ischaemic attack (TIA), and use of angiotensin-converting enzyme inhibitors or angiotensin receptor blockers, beta-blockers, diuretics, anticoagulants, antiplatelets, statins, mineralocorticoid receptor antagonists, and antiarrhythmics. We also controlled for time-varying covariates affected by prior exposure to AF: CIED-detected physical activity, AF burden, time-lagged AF burden and density in previous month, and a quadratic term for time (see Supplementary data online, Table S4). Cohort-specific adjustments were made based on data availability. In the VHA cohort, additional baseline covariates included systolic blood pressure and Care Assessment Need (CAN) score, with time-varying measures for beta-blockers, OAC, BMI, systolic blood pressure, and ejection fraction. In the UNC cohort, additional baseline covariates included prescriptions for sodium–glucose co-transporter 2 inhibitors.

To account for heterogeneity between study populations, cohort-specific RRs were pooled using random-effects meta-analysis.

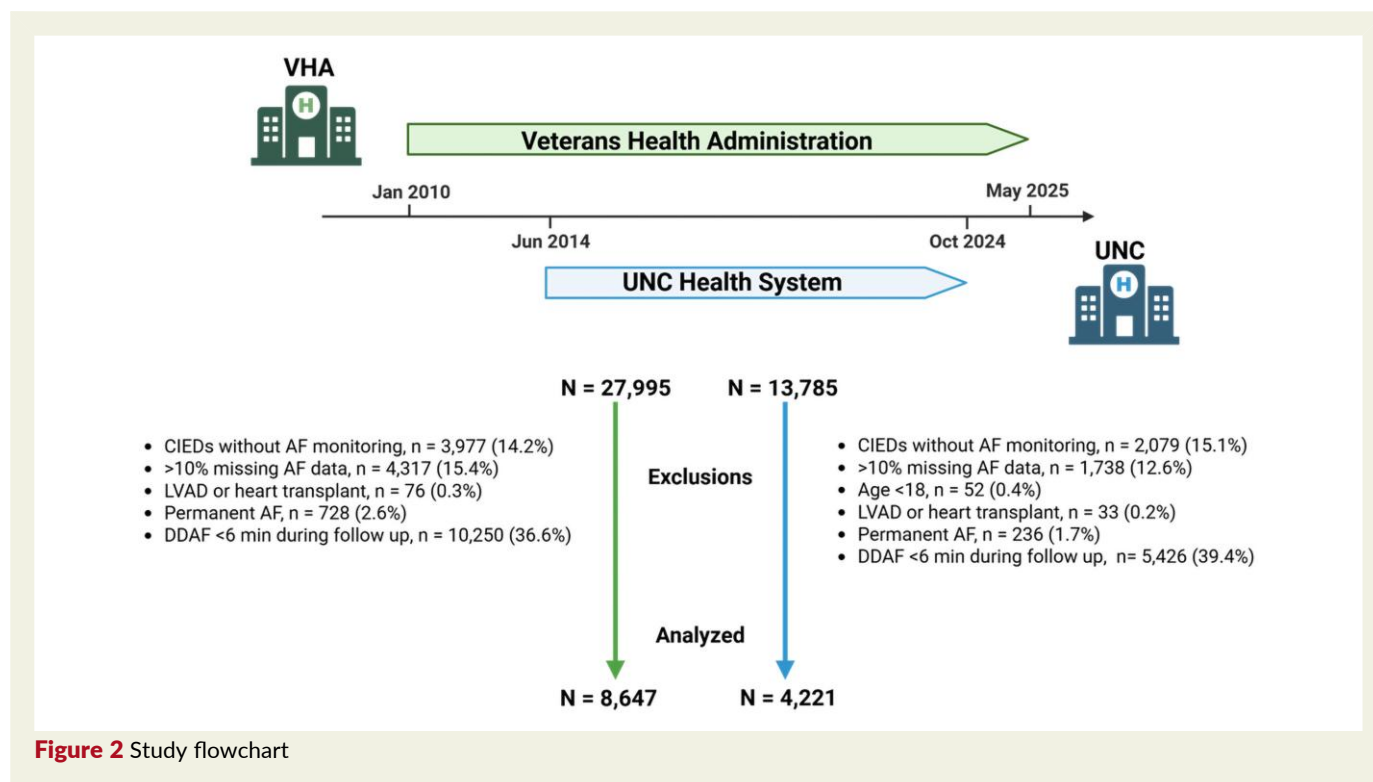


Figure 2 Study flowchart

Subgroup analyses were conducted across prespecified strata of baseline characteristics (e.g. device type, baseline diagnosis of AF or heart failure, history of ischaemic stroke or TIA, baseline OAC status, and age ≥ 75 vs < 75 years). A separate analysis stratified baseline diagnosis of AF by OAC status. To assess density's independent contribution beyond burden, we estimated stroke risk across all burden-density combinations using two complementary approaches: (i) within-stratum analyses comparing density levels within each burden category and (ii) joint analyses comparing all 16 combinations to a common reference of low density-low burden.

We conducted several sensitivity analyses to assess the robustness of our findings. First, to evaluate potential non-linearity in the exposure-outcome relationships, we used restricted cubic splines with knots at 0.3, 0.6, and 0.9 to model AF density and burden continuously across hypothetical 10% incremental AF density exposures (ranging from 0.1 to 1.0 in increments of 0.1). This approach ensured that our primary categorical contrasts accurately represented the underlying risk gradients. Second, to address potential exposure misclassification from gaps in CIED monitoring, we re-estimated risks under extreme-case scenarios assuming either 0% or 100% AF during missing days, thereby bounding the potential bias from incomplete data. Finally, to assess for g-null paradox and model misspecification, we compared observed and simulated means of time-varying covariates under the natural course. The fidelity of simulated trajectories was evaluated graphically to confirm that our parametric models adequately captured the data-generating process.

Results

Study population

Of the 41 780 patients with CIEDs enrolled in the VHA and UNC cohorts between 1 January 2010 and 30 May 2025, 12 868 (VHA: 8647; UNC: 4221) met our eligibility criteria (Figure 2).

Baseline characteristics are presented in Table 1. The mean age was 71.2 years in the VHA cohort and 73.7 years in the UNC cohort, with the UNC population including substantially more women (44.4% vs 2.3%) and a higher proportion of pacemakers (68% vs 45.4%). Both cohorts had median CHA₂DS₂-VASc score of 4, indicating a moderate risk of stroke, with high comorbidity burdens at baseline, including hypertension (90.7% VHA, 78.2% UNC), heart failure (53.6% VHA, 53.8% UNC), diabetes (50.0% VHA, 30.6% UNC), chronic kidney disease (32.8% VHA, 11.9% UNC), and prior ischaemic stroke or TIA (17.5% VHA, 10.2% UNC). Although the prevalence of clinical AF was comparable at baseline (59.3% VHA, 58.4% UNC), OAC rates differed markedly between cohorts (32.1% VHA vs 58.7% UNC). Among the baseline covariates, 11.2% of individuals in the VHA cohort had at least one missing baseline predictor (2.7% BMI, 2.5% systolic blood pressure, 8.1% left ventricular ejection fraction, and 6.2% CAN score). In the UNC cohort, only two individuals (<0.1%) were missing BMI data.

Atrial fibrillation density and stroke risk

During a median follow-up of 4.0 (IQR 1.9–6.1) years, ischaemic stroke occurred in 336 patients (2.6%): 251 (2.9%) in the VHA cohort and 85 (2.0%) in the UNC cohort. Among those who did not experience an ischaemic stroke during follow-up, 143 patients died of other causes (modelled as a competing risk), while the remaining patients were censored at the time of their last available CIED transmission. Among 202 784 person-months in which AF episodes were detected, 10.9% were classified as low density, 17.3% as medium density, 16.9% as medium-high density, and 54.9% as high density (see Supplementary data online, Figures S1 and S2). The observed and simulated 1-year stroke risks were 0.73% and 0.70% in

Table 1 Baseline characteristics of the cohorts stratified by the presence of clinical atrial fibrillation at the time of implant

	VHA		UNC	
	All patients (n = 8647)	Clinical AF Yes (n = 5131) No (n = 3516)	All patients (n = 4221)	Clinical AF Yes (n = 2463) No (n = 1758)
Demographics				
Age, years	71.2 (9.4)	71.9 (8.8)	73.7 (11.2)	71.5 (12.4)
Women	197 (2.3)	113 (2.2)	1876 (44.4)	716 (40.7)
Race				
White	7095 (82.1)	4295 (83.7)	3374 (79.9)	1325 (75.4)
Black	1039 (12.0)	543 (10.6)	709 (16.8)	364 (20.7)
Other/unknown	513 (5.9)	293 (5.7)	138 (3.3)	69 (3.9)
CHA ₂ DS ₂ -VASc	4.0 [3.0–5.0]	4.0 [3.0–5.0]	4.0 [3.0–5.0]	4.0 [2.0–5.0]
Device type and implant characteristics				
Manufacturer				
Medtronic	8647 (100.0)	5131 (100.0)	2292 (54.3)	1343 (54.5)
Boston Scientific	0 (0.0)	0 (0.0)	1596 (37.8)	662 (37.7)
Biotronik	0 (0.0)	0 (0.0)	333 (7.9)	147 (8.4)
Device type				
ICM	1279 (14.8)	791 (15.4)	0 (0.0)	0 (0.0)
Pacemaker	3930 (45.4)	2250 (43.9)	2872 (68.0)	1083 (61.6)
ICD	3438 (39.8)	2090 (40.7)	1349 (32.0)	675 (38.4)
Cardiac resynchronization therapy	1998 (23.1)	1242 (24.2)	963 (22.8)	413 (23.5)
Time since implant, years	4.2 [2.1–6.3]	3.8 [1.8–6.0]	3.6 [1.8–5.5]	4.0 [2.2–5.9]
Comorbidities				
Coronary artery disease	6141 (71.0)	3930 (76.6)	2294 (54.3)	1483 (60.2)
Chronic kidney disease	2838 (32.8)	1874 (36.5)	502 (11.9)	154 (8.8)
Diabetes	4327 (50.0)	2656 (51.8)	1293 (30.6)	444 (25.3)
Dyslipidaemia	7633 (88.3)	4699 (91.6)	2546 (60.3)	1690 (68.6)
Heart failure	4635 (53.6)	3152 (61.4)	2269 (53.8)	1452 (59.0)
Hypertension	7845 (90.7)	4835 (94.2)	3302 (78.2)	1175 (66.8)
Prior myocardial infarction	1717 (19.9)	1166 (22.7)	2119 (50.2)	737 (41.9)
Obstructive sleep apnoea	3061 (35.4)	2088 (40.7)	538 (12.7)	142 (8.1)

Continued

Table 1 Continued

	VHA		UNC	
	All patients (n = 8647)	Clinical AF Yes (n = 5131) No (n = 3516)	All patients (n = 4221)	Clinical AF Yes (n = 2463) No (n = 1758)
Valvular heart disease	2635 (30.5)	1810 (35.3)	1899 (45.0)	1352 (54.9)
Prior ischaemic stroke/TIA	1512 (17.5)	929 (18.1)	429 (10.2)	301 (12.2)
Procedures				
AF ablation	NA ^a	NA ^a	203 (4.8)	203 (8.2)
LAOO	NA ^a	NA ^a	20 (0.5)	20 (0.8)
PCI	724 (8.4)	472 (9.2)	469 (11.1)	296 (12.0)
CABG	570 (6.6)	405 (7.9)	940 (22.3)	635 (25.8)
Lifestyle factors				
Alcohol/substance-related disorder	2483 (28.7)	1538 (30.0)	286 (6.8)	164 (6.7)
Cigarette smoking ^b				
Current	2472 (28.6)	1402 (27.3)	293 (6.9)	142 (5.8)
Former	2890 (33.4)	1830 (35.7)	2009 (47.6)	1195 (48.5)
Never	2910 (33.7)	1773 (34.6)	1916 (45.4)	1125 (45.7)
BMI ^b , kg/m ²	30.5 (6.1)	30.6 (6.1)	28.5 (6.7)	28.4 (6.7)
Medications				
ACE-I/ARB	4529 (52.4)	2704 (52.7)	1937 (45.9)	1224 (49.7)
Beta-blocker	4734 (54.7)	3168 (61.7)	2894 (68.6)	1960 (79.6)
Calcium channel blocker	1789 (20.7)	1083 (21.1)	1220 (28.9)	858 (34.8)
Antiplatelet	2157 (24.9)	1201 (23.4)	1303 (30.9)	788 (32.0)
Anticoagulant	2779 (32.1)	2663 (51.9)	2477 (58.7)	1944 (78.9)
Statin	5232 (60.5)	3236 (63.1)	1867 (44.2)	1178 (47.8)
Antiarrhythmic	1245 (14.4)	1147 (22.4)	1227 (29.1)	1121 (45.5)
SGLT2i	74 (0.9)	53 (1.0)	201 (4.8)	115 (4.7)

Values are given as mean ± standard deviation, n (%), or median [interquartile range].

ACE-I, angiotensin-converting enzyme inhibitor; AF, atrial fibrillation; ARB, angiotensin receptor blocker; BMI, body mass index; CABG, coronary artery bypass graft; ICD, implantable cardiac defibrillator; ICM, insertable cardiac monitor; LAOO, left atrial appendage occlusion; NA, not available; PCI, percutaneous coronary intervention; SGLT2i, sodium-glucose co-transporter 2 inhibitor; TIA, transient ischaemic attack; VHA, Veterans Health Administration; UNC, University of North Carolina.

^aData not available from CMS per policy regarding cells with a value <10.

^bTwo hundred and thirty-two patients in the VHA cohort and 2 patients in the UNC cohort missing BMI and 375 patients in the VHA cohort and 2 patients in the UNC cohort missing cigarette smoking status.

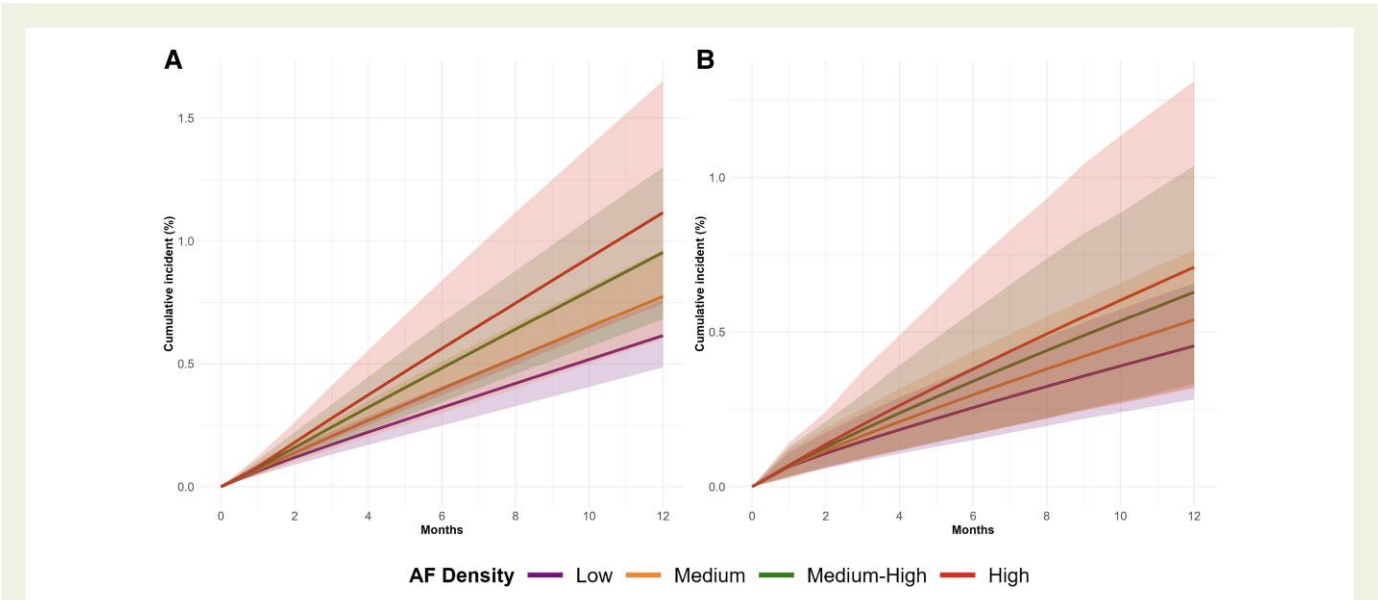


Figure 3 Adjusted cumulative incidence of ischaemic stroke at 1 year across increasing levels of atrial fibrillation (AF) density. The cumulative incidence of ischaemic stroke at 1 year are shown in the VHA cohort (A) and the UNC cohort (B)

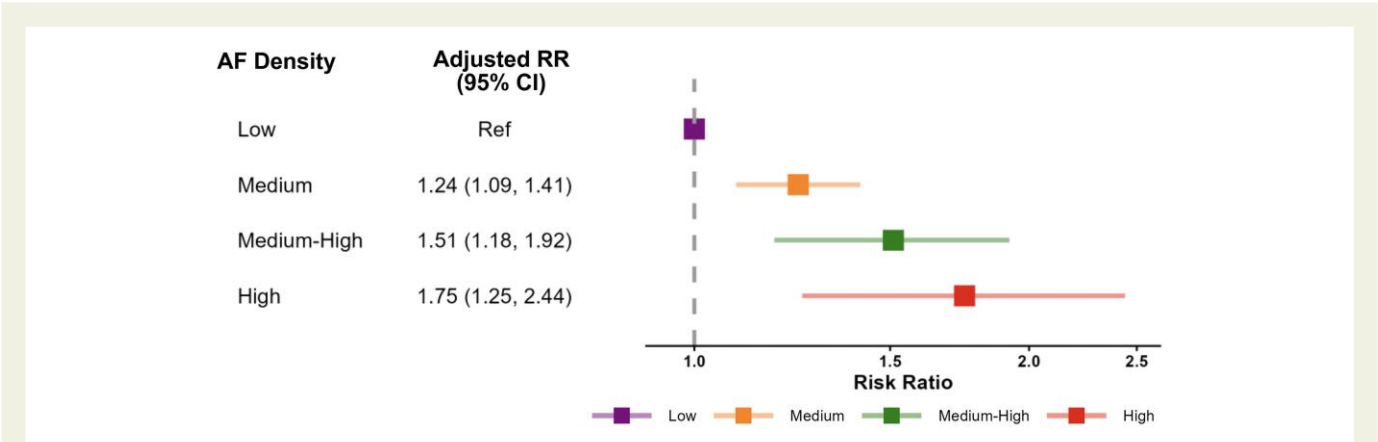


Figure 4 Adjusted 1-year risk of ischaemic stroke by levels of atrial fibrillation (AF) density. Forest plots depict the pooled risk ratios (RRs) for 1-year risks of ischaemic stroke by AF density category [low (>0–0.3), medium (>0.3–0.6), medium-high (>0.6–0.9), and high (>0.9–1.0)]. Estimates from parametric g-formula models adjusted for baseline (e.g. demographics, CHA₂DS₂-VAsC score, comorbidities, medications) and time-varying confounders (e.g. physical activity, anticoagulation) covariates, pooled using random-effects meta-analysis. Error bars represent 95% confidence intervals from 500 bootstrap samples

the VHA cohort, and 0.49% and 0.50% in the UNC cohort, respectively (see [Supplementary data online, Table S5](#)). There was a consistent dose–response relationship between increasing AF density and the 1-year risk of ischaemic stroke in both cohorts ([Figure 3](#)).

Stroke risk stratification with atrial fibrillation density

Pooled relative risks of ischaemic stroke by AF density level are shown in [Figure 4](#). Atrial fibrillation density demonstrated a robust dose–response relationship with 1-year stroke risk (see [Supplementary data online, Figure S3](#)). Relative to low density

(reference), risk increased progressively across categories: medium density (RR 1.24; 95% CI 1.09–1.41), medium-high density (RR 1.51; 95% CI 1.18–1.92), and high density (RR 1.75; 95% CI 1.25–2.44).

Subgroup analyses

Subgroup analyses demonstrated consistency with the main findings ([Figure 5](#); [Supplementary data online, Tables S6–S12](#)). Among patients without prior stroke, progressive risk increases were observed across density categories: medium (RR 1.29; 95% CI 1.09–1.51), medium-high (RR 1.61; 95% CI 1.19–2.20), and high (RR 1.91; 95% CI 1.25–2.91). Similar gradients were

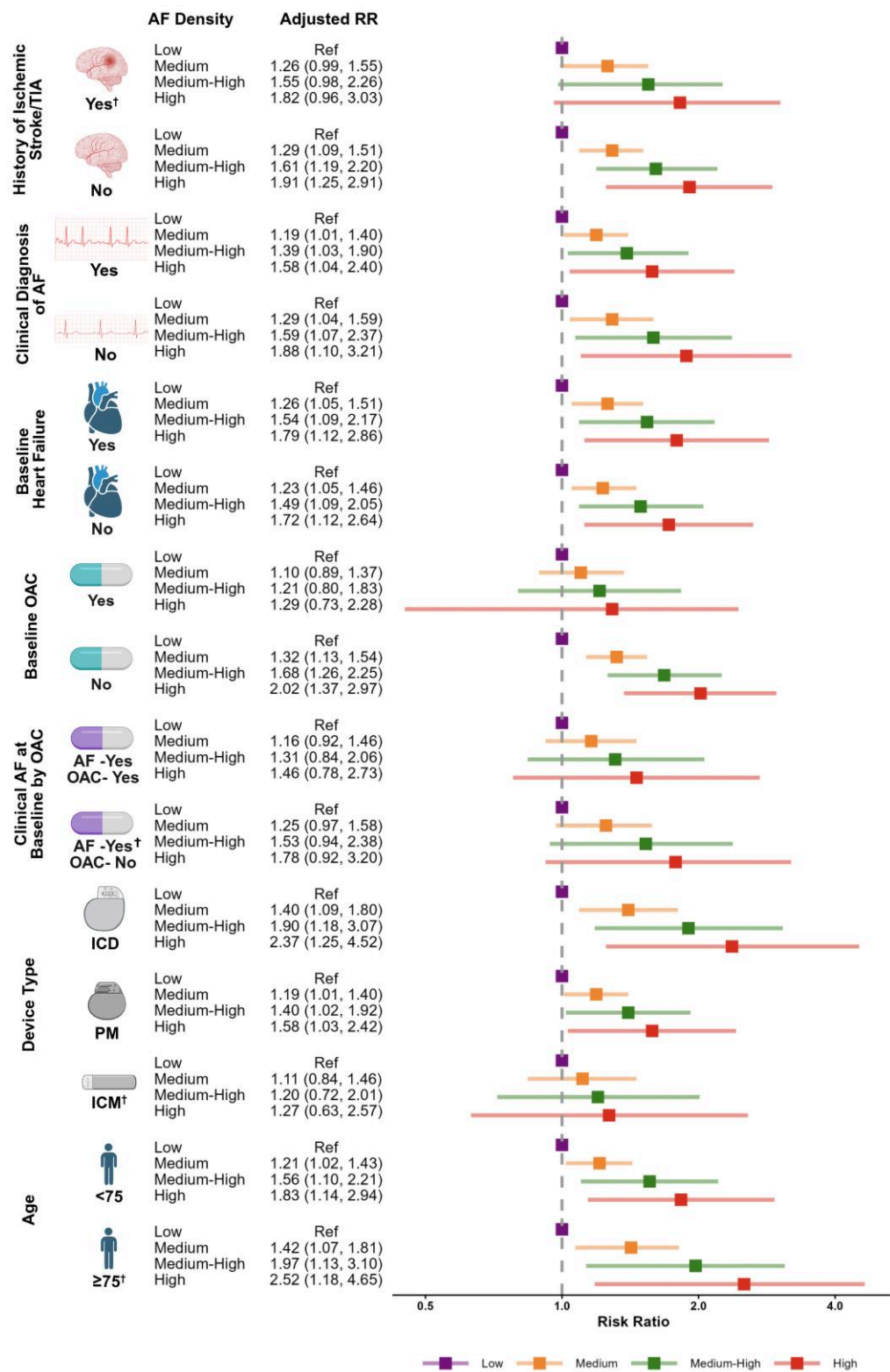


Figure 5 Adjusted 1-year risk of ischaemic stroke in prespecified subgroups. Forest plots depict the pooled risk ratios (RRs) for 1-year risks of ischaemic stroke by AF density category [low (>0–0.3), medium (>0.3–0.6), medium-high (>0.6–0.9), and high (>0.9–1.0)]. Estimates from parametric g-formula models adjusted for baseline (e.g. demographics, CHA₂DS₂-VAsC score, comorbidities, medications) and time-varying confounders (e.g. physical activity, anticoagulation) covariates, pooled using random-effects meta-analysis. Error bars represent 95% confidence intervals from 500 bootstrap samples. †Estimates from the VHA cohort only due to limited sample size in the UNC cohort

Table 2 Independent effect of atrial fibrillation density on stroke risk within atrial fibrillation burden strata

AF burden	AF density			
	Low	Medium	Medium-high	High
Low	1 (ref)	1.24 (1.09–1.42)	1.51 (1.18–1.94)	1.76 (1.25–2.47)
Medium	1 (ref)	1.21 (1.08–1.36)	1.42 (1.14–1.76)	1.61 (1.20–2.16)
Medium-high	1 (ref)	1.17 (1.06–1.29)	1.34 (1.11–1.63)	1.51 (1.16–1.96)
High	1 (ref)	1.16 (1.05–1.28)	1.32 (1.10–1.59)	1.48 (1.15–1.92)

RRs (95% confidence intervals) for 1-year ischaemic stroke by AF density within burden strata. Each row uses low density as reference. Within each burden level, increasing density is associated with progressively higher stroke risk, with the strongest gradient observed at lower burden levels. Estimates from parametric g-formula models pooled using random-effects meta-analysis.

Table 3 Stroke risk across all atrial fibrillation burden and density combinations

AF burden	AF density			
	Low	Medium	Medium-High	High
Low	1 (ref)	1.24 (1.09–1.42)	1.51 (1.18–1.94)	1.76 (1.25–2.47)
Medium	1.14 (1.03–1.27)	1.38 (1.22–1.56)	1.61 (1.33–1.94)	1.82 (1.40–2.36)
Medium-high	1.30 (1.00–1.67)	1.51 (1.21–1.89)	1.72 (1.39–2.15)	1.94 (1.51–2.49)
High	1.40 (0.95–2.05)	1.62 (1.14–2.30)	1.83 (1.36–2.46)	2.02 (1.51–2.71)

RRs (95% confidence intervals) for 1-year ischaemic stroke across burden-density combinations, with low burden-low density as a common reference. Reading across rows shows density effects; reading down columns shows burden effects. Both metrics independently increase risk, with patients exhibiting low burden but high density (RR 1.76) at greater risk than those with high burden but low density (RR 1.40). Estimates from parametric g-formula models pooled using random-effects meta-analysis.

AF, atrial fibrillation; RR, risk ratio.

observed in patients with prior cerebrovascular events, though with wider CIs.

Baseline anticoagulation status substantially modified the density–stroke relationship. Among untreated patients, density categories showed progressive increases: medium (RR 1.32; 95% CI 1.13–1.54), medium-high (RR 1.68; 95% CI 1.26–2.25), and high (RR 2.02; 95% CI 1.37–2.97). In contrast, associations were markedly attenuated among anticoagulated patients across all density levels. Among patients with clinical AF not receiving anticoagulation, medium-high density showed 53% increased risk (RR 1.53; 95% CI 0.94–2.38) and high density 78% increased risk (RR 1.78; 95% CI 0.92–3.20). Dose–response patterns persisted across device types. Among patients with ICDs, stroke risk increased from medium (RR 1.40; 95% CI 1.09–1.80) to high density (RR 2.37; 95% CI 1.25–4.52). Similar gradients were observed with pacemakers and ICMs. Additionally, age-stratified analyses revealed stronger associations in older patients. High density conferred 2.5-fold increased stroke risk among those aged ≥ 75 years (RR 2.52; 95% CI 1.18–4.65) compared with 1.8-fold increased risk in those < 75 years (RR 1.83; 95% CI 1.14–2.94).

Independent effects of atrial fibrillation density and burden on stroke risk

Stroke risk varied substantially among individuals with identical AF burden but different density patterns (Table 2). Within each

burden stratum, increasing density was associated with progressively higher stroke risk. Density's effect was strongest at lower burden levels, where high vs low density conferred 76% increased risk, with progressive attenuation at higher burden levels (48% increase in high burden stratum).

When examining all burden-density combinations relative to a common reference of low density-low burden (Table 3), both metrics were independently associated with increased stroke risk. The highest stroke risk was observed among individuals with high burden and high density (RR 2.02; 95% CI 1.51–2.71). Notably, patients with low burden but high density (RR 1.76; 95% CI 1.25–2.47) exhibited greater stroke risk than those with high burden but low density (RR 1.40; 95% CI 0.95–2.05).

Sensitivity analyses

Sensitivity analyses confirmed the primary findings. Results using restricted cubic splines demonstrated a consistent, dose–response relationship across incremental hypothetical AF density exposures, validating that our primary categorical thresholds did not misrepresent the risk gradient (see [Supplementary data online, Figure S3](#)). Extreme-case scenarios for missing CIED monitoring data (assuming 0% or 100% AF during gaps) yielded pooled RRs consistent with the primary analysis, successfully bounding the potential bias from exposure misclassification (see [Supplementary data online, Table S13](#)). The simulated and

observed risk, as well as time-varying covariate values, were closely aligned (see [Supplementary data online, Figures S4 and S5](#)), indicating that the risk of bias from the g-null paradox is minimal in this analysis.

Discussion

In this large, multicentre analysis of US patients with CIEDs, there was a robust dose-response relationship between increasing AF density and rates of ischaemic stroke at 1 year after extensive adjustment for established stroke risk factors, treatment, and AF burden. We also demonstrate that AF density provided superior stroke risk stratification compared with conventional burden metrics (i.e. percentage of time in AF), with consistent findings across device types, comorbidities, age groups, and anticoagulation status. These results suggest that temporal episode clustering provides clinically meaningful stroke risk information beyond cumulative arrhythmia burden. Most notably, patients with low burden but high-density AF exhibited higher stroke risk than those with high burden but low density AF, highlighting that the temporal pattern of AF episodes warrants consideration alongside traditional burden metrics in stroke risk assessment.

Contemporary risk stratification relies on clinical scoring systems such as CHA₂DS₂-VASc, which incorporate static patient characteristics but cannot capture dynamic AF patterns. This approach does not account for AF characteristics that may influence individual stroke risk, leaving many patients with an indication for chronic OAC. While continuous measures such as AF burden improve upon binary classification,³⁰ burden alone reduces temporal patterns to a single percentage (e.g. 20% burden from two consecutive days of sustained AF is treated identically to 20% burden from brief daily episodes), potentially obscuring clinically important differences in stroke risk.

To the best of our knowledge, this is the first study demonstrating that AF density provides prognostic information distinct from cumulative arrhythmia burden ([Structured Graphical Abstract](#)). Density's discriminative value was most pronounced at lower levels of AF burden and in populations where treatment decisions are most uncertain: patients without prior stroke and those with clinical AF not receiving anticoagulation. These findings directly address a major clinical challenge: anticoagulation decisions for low-burden AF, which remain controversial and lack clear guideline recommendations. Density could potentially identify which low-burden patients require treatment, thereby preventing strokes in high-risk patients who would otherwise remain untreated.

Atrial fibrillation density may also help explain heterogeneous findings from recent trials of anticoagulation in patients with device-detected atrial high-rate episodes (AHREs). NOAH-AFNET 6 found no benefit from edoxaban vs placebo for the combined endpoint of stroke and death [hazard ratio (HR) 0.81; 95% CI 0.60–1.08],³¹ while ARTESIA showed modest benefit from apixaban in reducing risk of stroke or systemic embolization (HR 0.63; 95% CI 0.45–0.88; absolute risk reduction 0.6%).³² Both trials enrolled patients based on AF burden thresholds, without considering temporal episode patterns. Patients with low density have lower baseline stroke risk, potentially diluting treatment effects. In contrast, individuals with high

density have a substantially higher baseline risk and may derive greater absolute benefit from anticoagulation. While the median duration of AF in both trials was low, unmeasured heterogeneity in temporal patterns may explain the divergent results. *Post hoc* analyses of both trials could test whether treatment effects were concentrated among high-density patients, potentially identifying subgroups that benefit from anticoagulation. Ongoing trials collecting AF burden data [REACT-AF (NCT05836987), CABA-HFPEF (NCT05508256), EASThigh-AFNET 11 (NCT06324188)] also provide immediate opportunities to prospectively test this hypothesis by incorporating density into patient selection or prespecified analyses.

The observation that patients with identical burden exhibit disparate densities with variable stroke risk suggests distinct pathophysiologic mechanisms. For instance, consolidated episodes of AF (high density) may promote sustained haemostatic activation, prolonged left atrial appendage stasis, and persistent endothelial dysfunction that collectively enhance thrombotic risk.^{23,33} Conversely, dispersed AF patterns (low density) with intervening sinus rhythm may facilitate haemostatic recovery, preventing prothrombotic perturbations from reaching critical thresholds for thrombus formation and propagation.³⁴ Further research correlating density with left atrial appendage flow dynamics and experimental models manipulating temporal patterns is needed to establish causality and identify therapeutic targets.

Critical research priorities include validating density across consumer and medical-grade wearables in diverse populations, investigating associations with other health outcomes (e.g. AF symptoms, heart failure, cognitive decline, quality of life, and mortality), and determining accuracy with shorter monitoring periods (<30 days) characteristic of intermittent ambulatory monitors.⁵ Prospective trials should also evaluate whether density-guided anticoagulation, particularly for intermediate-risk patients, improves outcomes compared with established clinical risk scores. Density should also be evaluated as both a stratification variable and therapeutic endpoint in rhythm control trials, potentially explaining heterogeneity in treatment responses.

If validated in future studies, AF density could fundamentally transform cardiac monitoring strategies and patient care. Current monitoring approaches focus on detecting episodes above arbitrary burden thresholds, yet findings from this investigation suggest that AF episode clustering may be more prognostically important. This is particularly relevant post-ablation, where recurrent AF episodes often exhibit high-density patterns despite lower burden, potentially explaining discordance between the rates of procedural success and stroke reported in the literature.³⁵ Implanted and wearable technologies could also leverage density for dynamic, real-time stroke risk assessment and facilitate personalized treatment decisions. For intermediate-risk patients with brief or infrequent AF episodes, where anticoagulation decisions are most uncertain, density patterns could provide the additional discrimination necessary for confident, shared decisions about when and how to initiate anticoagulation.

Major strengths of this study include the large sample of real-world patients from multiple US health systems with longitudinal CIED data from multiple manufacturers, enhancing generalizability. Our dynamic modelling approach also enabled

appropriate handling of time-varying exposures and confounders that traditional methods (i.e. Cox models) cannot accommodate. However, several limitations merit consideration. First, the observed stroke rate (6.3 per 1000 person-years) was lower than expected given the median CHA₂DS₂-VASC score of 4, though comparable to recent device-detected AF trials (NOAH-AFNET 6: 1.1% per year; ARTESIA: 1.0% per year). This likely reflects higher baseline anticoagulation use compared with these trials, which enrolled anticoagulation-naïve patients, and the fact that the CHA₂DS₂-VASC was derived and validated only in those with clinical AF. Despite lower event rates, robust associations between density and stroke suggest clinically meaningful risk discrimination even in partially treated populations. Second, stroke mechanism could not be determined and device-detected AF was not adjudicated, though manufacturer algorithms demonstrate high accuracy^{18–22} and professional societies in the USA and Europe have endorsed device-detected AF as a valid trial endpoint.^{5,36} Third, only dual-chamber pacemakers and ICDs with atrial leads were included in this study. Still, very brief or slow AF episodes may be missed by these devices.⁴ Fourth, our study focused on the first occurrence of ischaemic stroke after CIED implantation. While this is consistent with primary endpoints in major AF trials, the impact of AF density on stroke recurrence remains unknown and warrants future investigation using recurrent event modelling. Fifth, the validity of our causal estimates depends on parametric g-formula assumptions, including no unmeasured confounding (conditional exchangeability), no measurement error, and positivity. Residual confounding from unmeasured factors (e.g. autonomic tone, device indication) and outcome or covariate misclassification remain possible. Additionally, parametric g-methods are vulnerable to the g-null paradox, wherein model misspecification under the null can produce biased estimates. We mitigated this risk by comparing observed and simulated natural course trajectories, which demonstrated high fidelity across all covariates and outcomes (see [Supplementary data online, Table S5](#) and [Figures S3](#) and [S4](#)). Sixth, we used a random-effects (REML) meta-analysis rather than the Hartung–Knapp–Sidik–Jonkman (HKSJ) method.³⁷ With only two cohorts, HKSJ produces overly conservative CIs that preclude meaningful interpretation. Given our identical study protocols and consistent biological mechanisms across cohorts, REML provided an appropriate balance of precision and inference. Seventh, low event rates limited certain subgroup analyses and precluded stratification by sex and race, requiring further investigation in these populations. Finally, results may not generalize beyond patients with CIEDs, highlighting the need for further validation of density in other populations, technologies, and clinical settings.

Conclusions

Atrial fibrillation density represents a novel metric for stroke risk assessment, demonstrating that the temporal clustering of AF episodes provides clinically meaningful prognostic information beyond cumulative arrhythmia burden. While validation across other devices and populations, and demonstration of improved clinical outcomes through randomized trials remain essential, these findings establish a foundation for precision medicine

approaches that could enhance stroke prevention strategies and enable more personalized AF management.

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Supplementary data

Supplementary data are available at [European Heart Journal](#) online.

Declarations

Disclosure of Interest

L.R. has received research grants from NIH and Boston Scientific and served on a Biotronik advisory board. She also reports consultancy fees from Pfizer and Biotronik. S.S. is an employee and shareholder of Medtronic Inc. P.D.Z. is a contractor with and shareholder of Medtronic Inc. R.S.P. receives research support from the American Heart Association (18SFRN34250013) and the National Institutes of Health (UG3HL165065); advisory board fees from Medtronic, Abbott, Boston Scientific, CoreMap, and iRhythm; and royalties from UpToDate. K.W. has no relevant relationships to disclose. This publication does not necessarily represent the views of the Department of the Veterans Affairs or the United States Government.

Data Availability

Ethical and privacy reasons, as well as national health registry regulations, prohibit sharing of data. Further enquiries can be directed to the corresponding author.

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Ethical Approval

Institutional review boards at each site approved the study protocol with waiver of informed consent.

Pre-registered Clinical Trial Number

Not applicable.

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